

Combinations — Problem Set

Note: While the focus of this problem set is Combinations, questions may involve Permutations as well.

Warm-up

1. Express nC_r in terms of factorials.
2. In how many ways can a committee of 3 be chosen from 10 people?
3. A bag contains 5 red marbles and 4 blue marbles. In how many ways can 2 marbles be chosen if there are no restrictions?
4. From a group of 8 students, one captain and one vice-captain are to be chosen. In how many ways can this occur?
5. Simplify $\frac{n!}{(n-1)!}$.
6. A menu offers 4 starters and 6 main courses. How many different combinations of one starter and one main course are possible?
7. How many ways can a set of 3 books be selected from a shelf of 9 different books?
8. A basketball team of 5 players is chosen from a squad of 12. How many different teams are possible?
9. In how many ways can 4 people be selected from a group of 10 if one specific person must always be included?
10. Simplify the expression $\frac{(n+1)!}{n!}$.
11. A committee of 3 is chosen from 8 people. Find the number of ways this can be done.
12. A bag contains 5 red and 4 blue marbles. In how many ways can 3 marbles be chosen if there are no restrictions?

Skill-building

1. A group consists of 5 boys and 7 girls. In how many ways can a team of 4 be chosen if it must contain exactly 2 boys and 2 girls?
2. Solve for n : $\frac{(n+1)!}{n!} = 8$.
3. In how many ways can 6 people be seated around a circular table?
4. A committee of 5 is chosen from 6 doctors and 4 nurses. How many committees contain at least 4 doctors?
5. A box contains 10 light bulbs, 3 of which are defective. If 2 bulbs are chosen at random, find the probability that both are defective.

6. How many different 3-digit numbers can be formed using the digits 1, 3, 5, 7, 9 if no digit is repeated?
7. A committee of 4 is to be formed from 10 people. If one specific person must be included, how many ways can the committee be formed?
8. A committee of 5 is to be formed from 6 doctors and 4 nurses. How many committees can be formed if:
 - a) There are no restrictions?
 - b) The committee must contain exactly 3 doctors?
 - c) The committee must contain at least 4 nurses?
9. A group of 10 people includes Alice and Bob. How many ways can a committee of 4 be formed if:
 - a) Alice must be on the committee?
 - b) Alice and Bob refuse to serve together?
10. A volleyball team of 6 is chosen from 10 players. One player is designated as the captain. How many ways can this be done?
11. In how many ways can 3 letters be chosen from the word **COMBINE** and then arranged in a row?
12. There are 7 points on a circle. How many distinct triangles can be formed using these points as vertices?

Easier Exam Questions

1. (Gosford 2020 Q10a)
 - (a) Find the number of ways that 4 students can be chosen from 6 juniors and 3 seniors
 - (i) without restriction. 1
 - (ii) so that there are unequal numbers of juniors and seniors. 2
2. (Blacktown Boys 2020 Q12a)

From a group of 7 cats and 8 dogs, 10 are chosen at random as a pet package to go to a nursing home.

 - i) In how many ways can this pet package be chosen? 1
 - ii) In how many ways can this pet package be chosen if there are exactly 6 cats and 4 dogs? 1
 - iii) What is the probability that this pet package will consist of at least 6 cats? 2
3. (Caringbah 2019 Q8)

A school music band consists of 8 dancers and a singer.

The dancers are selected from 17 students in Year 9.

The singer is selected from 5 students in Year 10.

In how many ways could the band be selected?

- (A) ${}^{17}C_8 + {}^5C_1$
- (B) ${}^{17}P_8 + {}^5P_1$
- (C) ${}^{17}C_8 \times {}^5C_1$
- (D) ${}^{17}P_8 \times {}^5P_1$

4. (Blacktown Boys 2023 Q12a)

From a group of 9 girls and 5 boys, they are chosen to form different patrol groups of 5 to monitor a busy pedestrian crossing.

- (i) If there are no restrictions, how many different groups of 5 are possible? 1
- (ii) What is the probability of choosing a group with 3 girls and 2 boys? 2
- (iii) If a group of 3 girls and 2 boys stand in a line, what is the probability that the boys stand together? 2

5. (Baulkham Hills 2025 Q3)

In how many ways can 7 people be chosen from a group of 16 people and then arranged in a circle?

- (A) $\frac{15!}{9!}$
- (B) $\frac{15!}{9!7!}$
- (C) $\frac{16!}{9!}$
- (D) $\frac{16!}{9!7}$

6. (Baulkham Hills 2023 Q11c)

A team of 4 players consists of 2 men and 2 women. If a total of 7 players are available, of which 3 are men and 4 are women;

- (i) How many different teams can be selected? 1
- (ii) If 2 of the players are husband and wife, and wish to play on the same team, how many different teams can be selected if both are chosen? 1

7. (Baulkham Hills 2020 Q11d)

A group of 12 students is to be divided into discussion groups.

- (i) In how many ways can the discussion groups be formed if there are 7 people in one group and 5 people in the other? 1
- (ii) In how many ways can the discussion groups be formed if there are 3 groups containing 4 people each? 2
- (iii) In how many ways can a president and vice president be chosen for the group of 12? 1

8. (Baulkham Hills 2022 Q11d)

A committee of five is to be formed from a group of 5 men and 9 women. In how many ways can this be done if:

- (i) there are no restrictions? 1
- (ii) the committee consists of a majority of women? 2
- (iii) two of the men refuse to serve on the same committee? 2

9. (Knox 2022 Q7)

Out of 12 contestants, eight are to be selected for the final round of a competition. Five of those eight will be placed 1st, 2nd, 3rd, 4th and 5th.

In how many ways can this process be carried out?

- (A) $\frac{12!}{4!5!}$
- (B) $\frac{12!}{4!3!}$
- (C) $\frac{12!}{8!}$
- (D) $\frac{12!}{8!5!}$

10. (Hurlstone 2024 Q11b & c)

(b) Three integers are chosen at random from the integers 1 to 20 (inclusive) In how many ways can these integers be chosen if their sum is even? **2**

(c) Solve the equation $16(n-1)! = 5n! + (n+1)!$ where n is any positive integer. **3**

Harder Exam Questions

1. (Glenwood 2020 Q7a)

There are five women and six men in a group. From this group, a committee of four is to be chosen that contains three women and one man.

- i) How many different ways can this committee be formed? **1**
- ii) A particular man and woman must be in the committee. What is the probability that this committee is formed? **1**

2. (Sydney Tech 2020 Q9c)

A team of six students is to be formed from a class of eight students. How many different teams can be formed if two particular students cannot be both selected for the team? **1**

3. (SCEGGS 2021 Q5b)

In a class of 23 students, there are 12 girls and 11 boys.

- (i) The class selects a captain and vice-captain. In how many ways is this possible if there are no restrictions? **1**
- (ii) A group of 4 students consisting of 2 girls and 2 boys is chosen at random from the 23 students to go on the student representative council.
 - (α) In how many ways could this be done? **1**
 - (β) What is the probability that a certain girl, Alana, and a certain boy, Brian, have been included in the student representative council group? **2**

4. (Knox 2021 Q12b)

From a group of 6 men and 7 women, a committee of 5 people is to be formed.

- (i) How many committees can be formed if there are more men than women? **1**
- (ii) What is the probability that if the committee contains three men and two women, that only one of Anne or Harold is definitely in the committee? **1**

5. (Presbyterian 2021 Q6c)
A seven-person committee is to be chosen at random from 5 primary school students and 7 high school students.
- (i) In how many ways can this committee be chosen? 1
 - (ii) What is the probability that the committee will consist of exactly 4 high school students? 1
6. (Gosford 2022 Q3c)
A committee of 3 is to be elected from a club of 8 members:
- i. How many different committees can be formed? 1
 - ii. If there are 4 women in the club, what is the probability that a randomly selected committee of 3 contains only women? 2
7. (Knox 2023 Q11b)
From a group of 6 men and 7 women, a committee of 5 people is to be formed.
- (i) How many committees can be formed if there are more men than women? 1
 - (ii) What is the probability that if the committee contains three men and two women, that only one of Anne or Harold is definitely in the committee? 1
8. (Glenwood 2019 Q19)
There are 8 girls and 7 boys in a class.
- (a) The class elects a captain and a vice captain. In how many ways is this possible? 1
 - (b) The class elects four representatives for the student council. How many different groups of representatives are possible? 1
 - (c) How many groups of representatives for the student council are possible if the class decides that they want two girls and two boys in the group? 1
9. (North Sydney Boys 2023 Section C Q2)
Four digit numbers are formed from the digits 1, 2, 3 and 4. 1
Each digit is used once only. The sum of all the numbers that can be formed is?
- A) 6666
 - B) 44440
 - C) 66660
 - D) 266640
10. (Glenwood 2024 Q8e)
Solve the following where n is any positive integer: 2
- $$16(n-1)! = 5n! + (n+1)!$$
11. (Hurlstone 2019 Q8 c & d)
c) Twenty students, ten male and ten female, are travelling from school to the pool for the annual swimming carnival. Twelve of them go in a minibus, six of them on bikes and two of them walk.

- (i) Write a mathematical expression that will give the number of ways the students can be divided for the trip. **1**
- (ii) Show all working to give the numerical answer for the number of ways the students can be distributed, if none of the boys walk. **2**
- d) A committee of 5 members is to be selected at random from 5 Australians and 3 New Zealanders. Determine the probability that there will be a majority of Australians on the committee. Show all working out. **3**

12. (Girraween 2024 Q8 a,c & e)

- (a) Write $\frac{1}{(n+1)!} - \frac{1}{(n-1)!}$ as a single fraction **2**
- (c) A five-person team is to be chosen at random from eight women and nine men.
- i. In how many ways can this team be chosen? **1**
- ii. What is the probability that the team will consist of five men? **1**
- (e) In how many ways can a committee of 5 be selected from 10 people, consisting of 5 men and 5 women, if one man, Ken, refuses to work in the same committee with Sue? **2**

13. (Baulkham Hills 2024 Q14c)

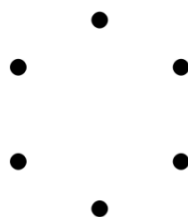
- (i) Show that ${}^{n+1}C_r - {}^nC_r = {}^nC_{r-1}$, for $r = 1, 2, 3, \dots, n$. **2**
- (ii) Hence find the value of ${}^3C_2 + {}^4C_2 + {}^5C_2 + \dots + {}^{100}C_2$. **2**

14. (Sydney Girls 2021 Q2d)

- 4 Maths books are chosen from 6 different Maths books and 3 English books are chosen from 5 different English books.
- (i) How many different ways can 7 books be selected? **1**
- (ii) Given that the 7 books have been selected, how many ways can they be arranged so that the Maths books stay together and the English books stay together? **1**

15. (Presbyterian 2023 Q14a)

- (i) Consider the dots below. **1**



Explain why 6C_3 distinct triangles can be formed using the dots as vertices.

- (ii) How many distinct triangles can be formed using the dots below as vertices? **2**

